

Navaneeth Maruthi

maruthi.n@northeastern.edu | (857)-421-9106 | LinkedIn | Github | Portfolio

PROFESSIONAL SUMMARY

Software engineer with 2+ years of production experience building distributed data systems and scalable backend infrastructure using **Apache Spark, Python, Snowflake, and AWS**. Hands-on expertise designing fault-tolerant pipelines, REST API-driven orchestration systems, and cloud-deployed applications. Experienced with **GenAI integration (OpenAI, Hugging Face, LLaMA 3.2)** and shipping full-stack products to real users.

EDUCATION

Northeastern University

MS in Computer Science

Visvesvaraya Technological University

BE in Computer Science and Engineering

Boston, MA

Expected Graduation: May 2027

Bangalore, India

May 2018 – July 2022

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6), SQL, C++, Go, C#

Frontend: React.js, Angular, HTML5, CSS3, Three.js

Backend: Node.js, Express.js, Spring Boot, REST APIs, Apache Kafka, Apache Spark

Databases: MySQL, MongoDB (Atlas), PostgreSQL, Snowflake, Hive, NoSQL

Cloud & DevOps: AWS (S3, ECS, ECR, SQS, Bedrock), Docker, Kubernetes, Terraform, GCP, CI/CD

AI & ML: OpenAI API, Hugging Face, LLaMA 3.2, Scikit-learn, PyTorch, Pandas, NumPy

Tools: Git, GitHub, Swagger, JWT, Agile/Scrum, Maven

WORK EXPERIENCE

Cognizant

Junior Software Engineer

Nov 2022 – Oct 2024

Bangalore, India

- Reduced data preparation time by ~40% by designing **Apache Spark-based ETL** jobs that migrated and transformed **10+ years** of historical datasets across **100+ dataset registrations** into the **AWS Refined Zone**, eliminating manual processing overhead for a **10-person** analytics team.
- Reduced new data source onboarding from **days to hours** by registering datasets and defining pipelines via a **Swagger-enabled** web interface, enabling dynamic configuration and reusability across **3 ingestion zones**.
- Managed data ingestion across a **3-zone architecture**: **Confirmed Zone** (raw validation & structural integrity), **Semantic Zone** (business logic & semantic transformations), and **Provisioning Zone** (downstream load preparation).
- Loaded **100+ datasets** into **Snowflake** for analytical reporting, enforcing **schema contracts** and data quality standards ensuring reliable consumption by a **10-person** enterprise reporting team.
- Leveraged **AWS S3** for intermediate staging under **Control-M** orchestration, validated end-to-end using **Prana** and **AWS Athena** — ensuring structural integrity across all datasets pre- and post-migration.
- Drove **100+ code reviews** in **Agile/CI/CD** sprints using **Git**, collaborating with engineers, QA, and business stakeholders to validate pipeline integrity and ship reliable software.

Cognizant

Software Engineer Intern

Apr 2022 – Sept 2022

Bangalore, India

- Developed a **Retail Billing Application** using **Spring Boot, Java, and H2 Database**, hosted on an **Angular** frontend as part of the GenC team under direct client supervision.
- Participated in **GenAI pilot initiatives**, implementing **OpenAI and Hugging Face**-based POCs to explore automation opportunities within internal enterprise workflows.
- Contributed to **AWS-hosted deployments** and supported integration of scalable **MLOps workflows** into internal tools.

PROJECTS

TradeFlow (GitHub) | Go, AWS SQS, ECS Fargate, Terraform

Spring 2026

- Built a distributed **FIX Protocol** order routing engine across **4 Go microservices** and **5 AWS SQS queues** on **ECS Fargate** with **Terraform IaC**; owned the **Order Router** and a **cost-aware circuit breaker** (Sony gobreaker) that trips on \$500/min slippage — processed **79,122 orders** with **zero failures** at peak **193 req/sec**.

BodyMap 3D (GitHub) | React, Three.js, Node.js, MongoDB

June 2026

- Built a full-stack health symptom tracker with **React** and **Three.js** (React Three Fiber) featuring raycasting-based 3D body region detection, severity-coded pain markers, a real-time analytics dashboard, and **AI health insights** via a locally hosted **LLaMA 3.2** model; secured with **JWT** auth and bcrypt.

Alignify (Live Demo) | React, Node.js, Express, MongoDB

Nov 2025

- Architected and shipped a full-stack goal-planning app enabling **50+ users** to convert long-term goals into **1,398 structured tasks** (daily, weekly, quarterly) with **JWT authentication**, protected routes, **Docker** containerization, and **WCAG-compliant** design.